

# TCForge: physics-matched synthesis from measured forward model

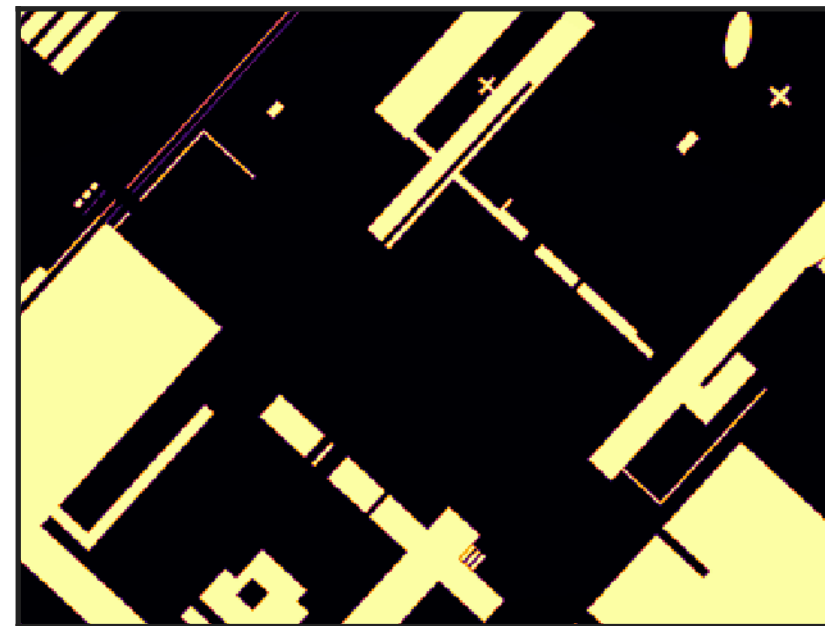
# TCForge: physics-matched synthesis from measured forward model

## 1. Scene geometry



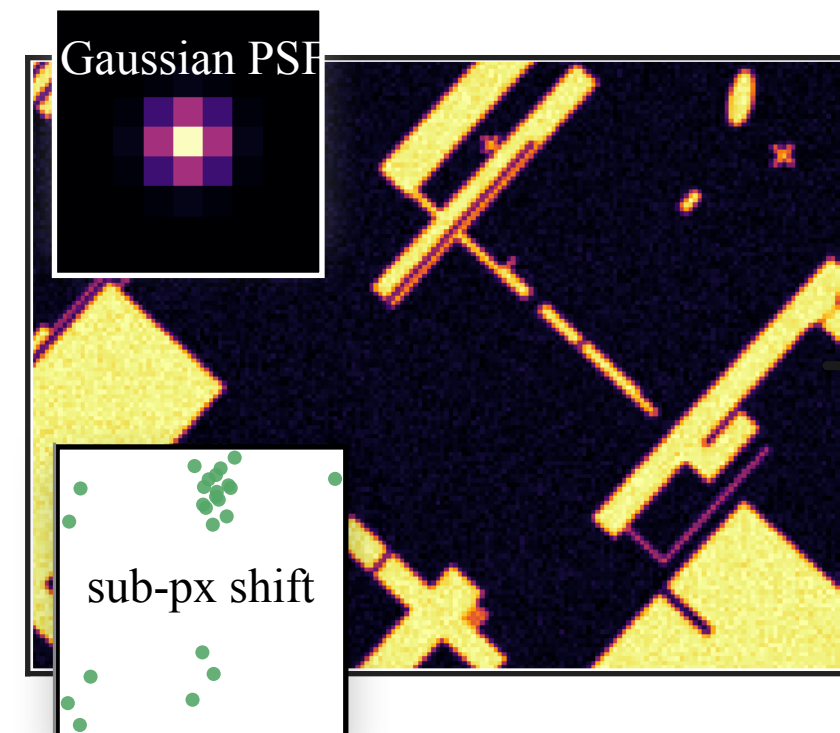
8-layer IC primitives  
rotate  $\theta = 47.6^\circ \pm 1.5^\circ$   
4x SSAA  $\rightarrow$  coverage  $\in [0, 1]$

## 2. Temperature field



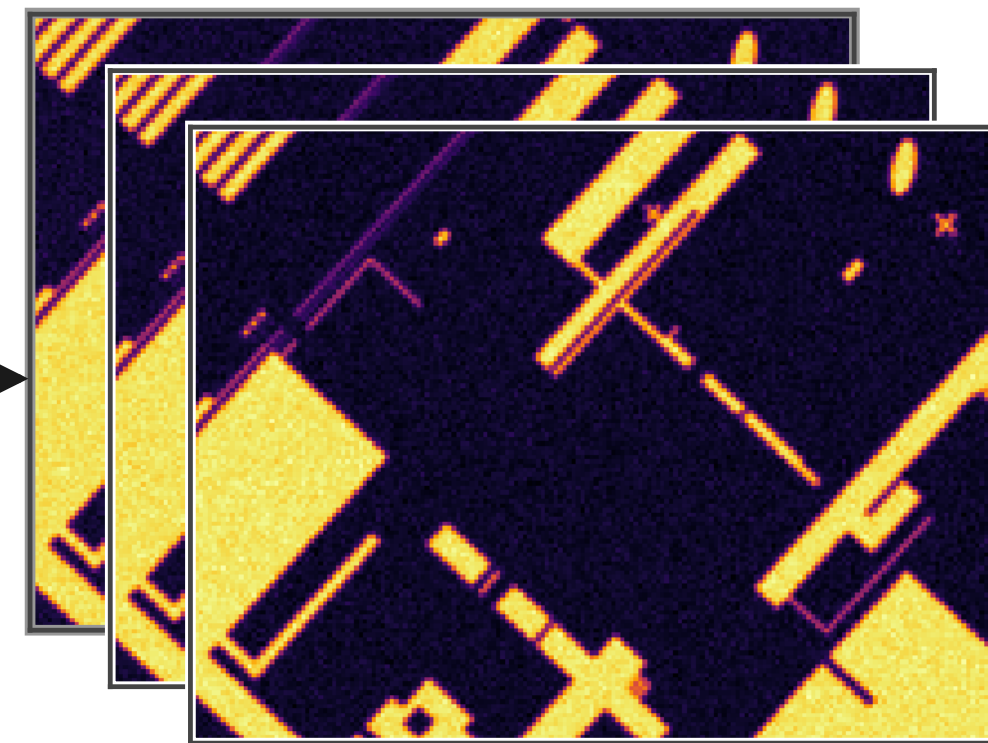
$T = T_{bg} + \Delta T \cdot cov + drift$   
 $\Delta T \sim U(0.5, 5.0)^\circ C$   
smooth low-freq drift

## 3. Forward degradation



Gaussian PSF  $\sigma \sim U(0.15, 0.55)$   
 $\rightarrow$  block-avg downsample ( $\times 2$ )  
sub-pixel shifts (248 profile)  
+ 0.02px jitter, noise  $0.0724^\circ C$

## 4. LR burst $\rightarrow$ training pair



1000 scenes  $480 \times 640 \rightarrow 960 \times 1280$   
hybrid input:  $K = 4$  drizzle variants  
(LR burst, shifts) paired to HR

1x Stats (5ch)  
Hybrid Drizzle (8ch)

Input  
Selector

